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Growth Mixture Models in the Structural Equation Modeling Versus the Multilevel Modeling Framework

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ABSTRACT

The performance of linear growth mixture models (GMMs) estimated in the multilevel modeling (MLM) framework and the structural equation modeling (SEM) framework in Mplus is examined. Empirical analyses conducted using data from the Early Childhood Longitudinal Study – Kindergarten Class of 1998–1999 (ECLS-K) led to different estimates of a 2-class linear GMM when estimated in the MLM versus the SEM frameworks via Mplus. A simulation study was conducted under varying conditions of sample size and class separation to better understand when differences in model fit and parameter estimates emerge. Results suggested the frameworks yielded comparable model fit information and parameter estimates when class separation was large; however, differences emerged as class separation decreased with the MLM estimation routine tending to outperform the SEM estimation routine. Overall, these findings highlight the importance of considering the estimation framework when estimating linear GMMs in Mplus.

KEYWORDS

Developmental trajectories;
finite mixture modeling;
latent class analysis;
longitudinal growth
modeling; multilevel
modeling

1. Introduction

Developmental researchers focus on behavioral change and many of their research questions are best investigated through the collection and analysis of longitudinal data. Longitudinal methods include a wide range of statistical models to track individual changes over time, including models to analyze panel data in sociology, time series data in econometrics, and single-subject data in operant psychology.

Growth models (Bryk & Raudenbush, 1987; Laird & Ware, 1982; Meredith & Tisak, 1990) are commonly used in the social, behavioral, and educational sciences to model individual changes (growth, decline) and account for variation in how individuals change over time. Growth models are often specified and estimated in the multilevel (MLM; long data format) or structural equation modeling (SEM; wide data format) framework. While both frameworks have relative advantages and disadvantages for the specification of linear growth models (Lindenberger & Ghisletta, 2004), the MLM and SEM frameworks yield identical or near identical parameter estimates for comparably specified linear growth models. This makes the choice of modeling framework a choice of convenience or familiarity.

An extension of the growth model, the growth mixture model (GMM; Muthén & Shedden, 1999), is an exploratory approach to search for groups of individuals with similar change patterns. Like the growth model, GMMs can be specified in the MLM framework or the SEM framework. The SEM framework is most often utilized to estimate GMMs because Mplus (Muthén & Muthén, 1998–2012) began as an SEM program that had the ability to model mixture distributions.

Mplus added the ability to estimate MLMs in version 2 (MLMs with random slopes in version 3) and MLM mixture models in version 4.2; however, its use remained limited.

While the equivalence of the parameter estimates and model fit statistics for growth models fit in the MLM and SEM frameworks has been well documented (Curran, 2003; Willett & Sayer, 1994), the same has not been examined for GMMs. Moreover, the estimation of GMMs is more complicated than the estimation of growth models as local solutions are common (Hipp & Bauer, 2006). In this paper, we discuss growth models and GMMs, describe their specifications in the MLM and SEM frameworks, examine discrepancies in parameter estimates when analyzing empirical data, and conduct simulations to compare how well these frameworks recover population values of linear GMMs.

2. Growth Modeling

Growth models are a collection of longitudinal models to decompose repeated measures data into an underlying smooth trajectory and noise. The smooth trajectory is a parametric function, and the parameters of the function are allowed to vary across individuals. This allows individuals to have differing rates of change. For example, if a linear trajectory is specified, each individual has an intercept, representing their expected value at a specific point in time, and a slope, representing their rate of change. Instead of estimating an intercept and slope for each individual, growth models assume a distributional form for the growth parameters. The multivariate normal distribution is commonly assumed

with estimated means, variances, and covariances of the growth parameters. Given the multivariate normality assumption of the growth parameters, growth models essentially assume that the sample is drawn from a single population.

2.1. Multilevel Modeling Framework

In the MLM framework, growth models are two-level models. Within-person change is captured by the level-1 model, and between-person differences are captured by the level-2 model. The linear growth model is the most commonly specified growth model, so we discuss this model throughout the paper. As an MLM, the linear growth model is written as

$$y_{ti} = b_{0i} + b_{1i} \cdot x_{ti} + u_{ti} \quad (1)$$

where, y_{ti} is the repeatedly measured score at assessment t for individual i , x_{ti} is the value of the time metric at assessment t for individual i , b_{0i} is the random intercept for the individual i and the expected value of y_{ti} when $x_{ti} = 0$ for individual i , b_{1i} is the random slope and expected change in y_{ti} for a one-unit change in x_{ti} for individual i , and u_{ti} is the residual at assessment t for individual i . The residuals are assumed to follow a normal distribution with a time-invariant variance (i.e., $u_{ti} \sim N(0, \sigma_u^2)$).

The random intercept and slope are written as deviations from their population means in the level-2 model, such that

$$\begin{aligned} b_{0i} &= \beta_0 + d_{0i} \\ b_{1i} &= \beta_1 + d_{1i} \end{aligned} \quad (2)$$

where β_0 and β_1 are the sample means for the intercept and slope, and d_{0i} and d_{1i} are individual i 's deviation from the sample mean of the intercept and slope. These individual deviations are assumed to follow a multivariate normal distribution (i.e., $d_{0i}, d_{1i} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{10} \\ \sigma_{10} & \sigma_1^2 \end{bmatrix}\right)$).

2.2. Structural Equation Modeling Framework

In the SEM framework, growth models are specified as restricted confirmatory factor models where the growth factors are latent variables (i.e., common factors). With the data in the wide format, the linear growth model is written in matrix form as

$$\mathbf{y}_i = \mathbf{\Lambda} \boldsymbol{\eta}_i + \mathbf{u}_i \quad (3)$$

where $\mathbf{y}_i$ is a $T \times 1$ vector of the repeatedly measured scores for the individual i , where T represents the number of repeated assessments based on the timing metric, $\mathbf{\Lambda}$ is a $T \times R$ matrix of factor loadings defining the growth factors (latent variables), where R is the number of growth factors ($R = 2$ for linear growth), $\boldsymbol{\eta}_i$ is an $R \times 1$ vector of latent factor scores for the individual i , and $\mathbf{u}_i$ is a $T \times 1$ vector of residual scores for the individual i . As in the MLM, the latent factor scores are written as deviations from the sample means, such that

$$\boldsymbol{\eta}_i = \boldsymbol{\alpha} + \boldsymbol{\xi}_i \quad (4)$$

where $\boldsymbol{\alpha}$ is an $R \times 1$ vector of factor means and $\boldsymbol{\xi}_i$ is an $R \times 1$ vector of mean deviations for i . The covariance matrix for $\boldsymbol{\xi}_i$, $\boldsymbol{\Psi}$, has freely estimated elements.

2.3. Growth Modeling in Mplus

Growth models are often fit in the MLM framework or the SEM framework in Mplus using a maximum likelihood (ML) estimator. In the MLM framework, the growth parameters are *random coefficients* and the data are organized in the long format, where there are multiple records for each individual. In Mplus, ML with robust standard errors (i.e., MLR) is the default estimator for MLM specified growth models and numerical integration with quadrature is used to approximate the likelihood function. In the SEM framework, the growth parameters are *latent variables* and the data are organized in the wide format, where there is a single record for each individual with separate variables for each repeatedly measured variable. Here, Mplus uses ML estimation and estimates parameters that minimize differences between the observed and expected covariance matrix and mean vector.

2.4. Advantages and Disadvantages

Willett and Sayer (1994) highlighted the fundamental mathematical equivalence of the SEM and MLM frameworks for the specification and estimation of growth models. When estimated, the SEM and MLM frameworks yield identical or near identical parameter estimates. Empirical and analytical examples from Curran (2003) showed that two-level models estimated in the SEM and MLM frameworks led to the same results and parameter estimates to the third decimal place.

As we noted, there are some relative advantages and disadvantages to specifying growth models in each framework. Some of the advantages and disadvantages to using each framework arise with differences in data structure, model complexity, and model fit information (Lindenberger & Ghisletta, 2004). Data are structured differently in each framework as it relates to the timing variable. In the MLM framework, the timing variable (i.e., x_{ti}) is included as a predictor variable, whereas in the SEM framework, the timing variable is specified as a set of parameters in the factor loading matrix. This difference makes handling unbalanced or incomplete data easier in the MLM framework, where unbalanced data are when individuals are measured at various time points or data where some individuals have fewer observations than others.

The wide data setup in the SEM framework permits the comparison of expected and observed covariance matrices in order to obtain the chi square statistic and related model fit indices. This allows researchers to calculate model fit statistics (e.g., Root Mean Square Error of Approximation [RMSEA], Comparative Fit Index [CFI], and Tucker-Lewis Index [TLI]) that quantify how well the model fits the data.

In the MLM framework, the inclusion of time as a predictor limits model fit indices to likelihood-based fit statistics (e.g., $-2 \log$ -likelihood $[-2LL]$, Bayesian Information Criterion [BIC], Akaike Information Criterion [AIC]), which allow researchers to compare models but do not allow researchers to examine how well the model represents the data.

3. Growth Mixture Modeling

GMMs extend the growth model to multiple latent populations, where a latent categorical variable separates the sample into multiple classes (groups of individuals). The GMM can be thought of as an extension of a multiple group growth model, where growth models are specified for each measured group, and the equivalence of their parameters (i.e., means, variances, and covariances) is examined statistically (see Ram & Grimm, 2009). The unobserved classes in GMMs are inferred from the data and identified in a *post hoc* fashion. Researchers using GMMs aim to uncover information about these latent classes, including the number of latent classes, the class-specific growth trajectories, and the sample sizes for each latent class.

3.1. Multilevel Modeling Framework

In the MLM framework, a linear GMM is a collection of two-level models (one for each class), where the repeated measures are nested within the individual. Within each class, the level-1 model captures within-person change and the level-2 model captures between-person differences within each latent class. As an MLM, the linear GMM is written as

$$y_{ti}^{(k)} = b_{0i}^{(k)} + b_{1i}^{(k)} \cdot t + u_{ti}^{(k)} \quad (5)$$

where $y_{ti}^{(k)}$ is the repeatedly measured score at assessment t for individual i , $b_{0i}^{(k)}$ is the random intercept and expected value of y_{ti} when $t = 0$ for individual i in class k , $b_{1i}^{(k)}$ is the random slope and expected change in y_{ti} for a one-unit change in t for individual i in class k , and u_{ti} is the residual at assessment t for individual i in class k . The residuals are assumed to follow a normal distribution with a time-invariant variance for the class k (i.e., $u_{ti}^{(k)} \sim N(0, \sigma_u^{2(k)})$).

The random intercept and slope are written as deviations from their class-specific population means, such that

$$\begin{aligned} b_{10}^{(k)} &= \beta_0^{(k)} + d_{0i}^{(k)} \\ b_{1i}^{(k)} &= \beta_1^{(k)} + d_{1i}^{(k)} \end{aligned} \quad (6)$$

where $\beta_1^{(k)}$ and $\beta_2^{(k)}$ are the class-specific sample means for the intercept and slope, and $d_{1i}^{(k)}$ and $d_{2i}^{(k)}$ are the individual deviations for the individual i in class k from the respective class-specific sample mean. Individual deviations are assumed to follow a multivariate normal distribution for the class k (i.e., $d_{10}^{(k)}, d_{1i}^{(k)} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^{2(k)} & 0 \\ 0 & \sigma_1^{2(k)} \end{bmatrix}\right)$). This specification allows for all of the model's parameters to be

class specific. While this is possible, researchers often focus their attention on the means of the latent classes ($\beta_1^{(k)}$ and $\beta_2^{(k)}$) and only allow these parameters to differ across classes.

3.2. Structural Equation Modeling Framework

In the SEM framework, linear GMMs are specified as restricted confirmatory factor mixture models. As in the MLM framework, a class-specific designation is given for most aspects of the model. The linear GMM is written as

$$y_i^{(k)} = \Lambda \eta_i^{(k)} + u_i^{(k)} \quad (7)$$

where $y_i^{(k)}$ is a $T \times 1$ vector of the repeatedly measured scores for the individual i in class k , Λ is a $T \times 2$ A matrix of factor loadings defining the growth factors (latent variables) and contains fixed values because this is a *linear* GMM, $\eta_i^{(k)}$ is a 2×1 vector of latent factor scores for the individual i in class k , and $u_i^{(k)}$ is a $T \times 1$ A vector of unique scores for an individual i in class k . The latent factor scores are written as deviations from the class-specific sample means, such that

$$\eta_i^{(k)} = \alpha^{(k)} + \xi_i^{(k)} \quad (8)$$

where $\alpha^{(k)}$ is an 2×1 vector of latent factor means for class k and $\xi_i^{(k)}$ is an 2×1 vector of residual deviations for individual i in class k . The covariance matrix for $\xi_i^{(k)}$ is $\Psi^{(k)}$ and has freely estimated elements for the class k .

3.3. Growth Mixture Modeling in Mplus

GMMs are estimated in the MLM framework in Mplus with the MLR estimator with the Expectation-Maximization (EM) algorithm and numerical integration. The EM algorithm is used to iteratively determine latent class membership and numerical integration is used to approximate the likelihood function. GMMs specified in the SEM framework in Mplus are estimated with MLR estimation by default with the EM algorithm to maximize the likelihood, determine optimal parameter estimates and latent class membership.

4. Application to Empirical Data

Grimm et al. (2017) discussed linear GMMs and fit a series of linear GMMs to empirical data from the Early Childhood Longitudinal Study – Kindergarten Class of 1998–1999 (ECLS-K). The outcome was the child's weight, which was measured in the fall and spring of kindergarten and first grade. The 2-class model with the means of the intercept and slope allowed to vary over classes was considered the best representation of the growth trajectories. This model was fit in the SEM framework. These data were reanalyzed with a comparably specified 2-class GMM in the MLM framework in Mplus. To aid consistency in the parameter estimates, we provided equivalent starting values for the model's parameters. Programming scripts for these models are contained on the second author's website.

Table 1. Model fit statistics and parameter estimates from the 2-class linear growth mixture model fit to the ECLS-K data in the SEM and MLM frameworks.

	Structural Equation Model		Multilevel Model	
Model Fit Statistics				
–2LL	23,601		23,590	
AIC	23,619		23,608	
BIC	23,664		23,665	
SABIC	23,635		23,637	
Parameter Estimates				
	Class 1 <i>N</i> = 994	Class 2 <i>N</i> = 89	Class 1 <i>N</i> = 973	Class 2 <i>N</i> = 110
Intercept Mean	44.06 (0.23)	64.31 (2.52)	44.34 (0.04)	60.25 (0.39)
Slope Mean	5.82 (0.13)	15.30 (0.70)	5.70 (0.10)	14.85 (0.57)
Intercept Variance	40.08 (2.74)	40.08 (2.74)	43.75 (3.55)	43.75 (3.55)
Slope Variance	3.12 (1.35)	3.12 (1.35)	2.55 (1.82)	2.55 (1.82)
Intercept-Slope Covariance	6.69 (1.21)	6.69 (1.21)	6.30 (1.50)	6.30 (1.50)
Residual Variances	5.57 (1.34)	5.57 (1.34)	5.18 (1.76)	5.18 (1.76)

Note: Standard errors are contained within parentheses.

The parameter estimates and model fit statistics for these models are contained in Table 1. The results indicate that different solutions were identified as the model fit statistics and the parameter estimates were notably different between frameworks. In the SEM framework, the first class held $n = 994$ participants with a mean intercept equal to 44.06 and a mean slope equal to 5.82. The second class held $n = 89$ participants with a mean intercept equal to 64.31 and a mean slope equal to 15.30. In the MLM framework, the first class held $n = 973$ participants with a mean intercept equal to 44.34 and a mean slope equal to 5.70. The second class held $n = 110$ participants with a mean intercept equal to 60.25 and a mean slope equal to 14.85. The differences in the model fit and the parameters for each class suggest research is needed to determine when model fit and parameter estimates align and when they do not align between the SEM and MLM frameworks for linear GMMs, and whether the estimates from one framework better capture true underlying class differences.

5. Research Questions

As shown, parameter estimates and latent class sizes can vary when linear GMMs are estimated in the SEM and the MLM frameworks in Mplus. Next, we use simulation to determine when these frameworks recover equivalent parameter estimates (class means, class-specific growth parameters, etc.) and lead to the same conclusions, as is the case with linear growth models, and when the estimates differ between frameworks. Furthermore, we examine whether one framework better captures the true underlying class differences.

6. Method

6.1. Simulation

We simulated data from a two-group linear growth model and then fit comparably specified growth and GMMs in both frameworks to examine model fit and parameter recovery for the 2-class GMM. The linear growth data for each

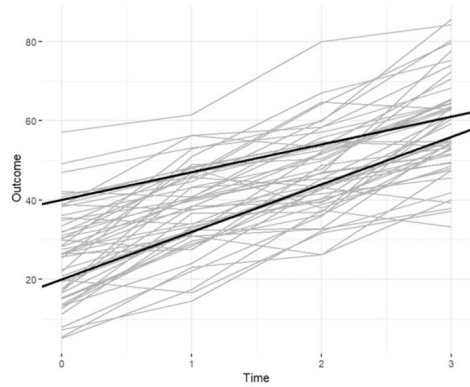
group were simulated with normally distributed growth factors (intercept and slope) and normally distributed residual terms within each class and four assessments (coded as 0, 1, 2, and 3). The growth trajectories of the two groups only differed in the means of the growth factors.

Multiple samples were drawn from the population with varying sample size and group separation, as the group separation is considered a primary determinant of class identification and recovery in GMMs (Lubke & Muthén, 2007; Nylund et al., 2007). Sample sizes were 500, 1000, 1500, and 2000 with 60% of the sample coming from the first group and 40% of the sample coming from the second group. The random-effects parameters for each group were $\sigma_0^2 = 64$, $\sigma_1^2 = 4$, $\sigma_{10} = 0$, and $\sigma_e^2 = 25$. There were five levels of class separation (2.5 SD, 2.0 SD, 1.5 SD, 1.0 SD, and 0.5 SD). In the smallest group separation case, the intercept and slope means for the two classes were 0.5 standard deviations apart, and in the largest case, the intercept and slope means were 2.5 standard deviations apart. In the 2.5 SD condition, the means of the intercept and slope were $\beta_0 = 40$ and $\beta_1 = 7$ for the first group and $\beta_0 = 20$ and $\beta_1 = 12$ for the second group. In the 2.0 SD condition, the means of the intercept and slope were $\beta_0 = 38$ and $\beta_1 = 7.5$ for the first group and $\beta_0 = 22$ and $\beta_1 = 11.5$ for the second group. In the 1.5 SD condition, the means of the intercept and slope were $\beta_0 = 36$ and $\beta_1 = 8$ for the first group and $\beta_0 = 24$ and $\beta_1 = 11$ for the second group. In the 1.0 SD condition, the means of the intercept and slope were $\beta_0 = 34$ and $\beta_1 = 8.5$ for the first group and $\beta_0 = 26$ and $\beta_1 = 10.5$ for the second group. In the 0.5 SD condition, the means of the intercept and slope were $\beta_0 = 32$ and $\beta_1 = 9$ for the first group and $\beta_0 = 28$ and $\beta_1 = 10$ for the second group. The group separation is visualized in Figure 1, where a sample of individual trajectories from both classes with the average growth trajectory is plotted with differing levels of class separation.

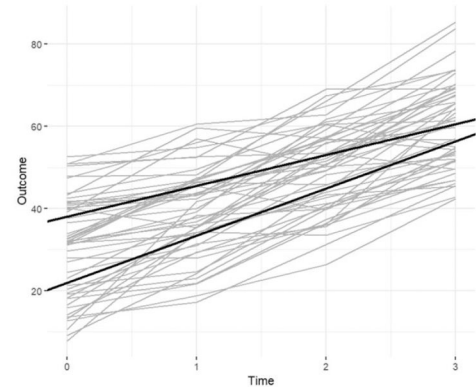
6.2. Model Generation and Estimation

Data were generated in R (R Core Team, 2023) and the MplusAutomation (Hallquist & Wiley, 2018) package facilitated communication between R and Mplus. Three models were fit to each simulated dataset. First, a linear growth model was fit to compare the log-likelihood values across frameworks to highlight the equivalence of the frameworks for estimating growth models. Second, a 2-class GMM with means allowed to vary over latent classes was fit. This GMM matched the data generation process as there were two groups in the data and the two groups only differed in the means of the intercept and slope. The model fit information and the parameter estimates were output from these models to examine differences across frameworks and examine parameter bias. Finally, a 3-class GMM with means allowed to vary over latent classes was specified. The fit of the linear growth model, the 2-class GMM, and the 3-class GMM were compared for model selection. Specifically, we compared the sample size adjusted BIC (SABIC) for these models within each framework.

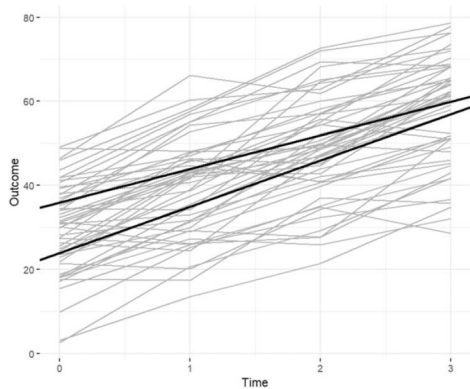
(A)



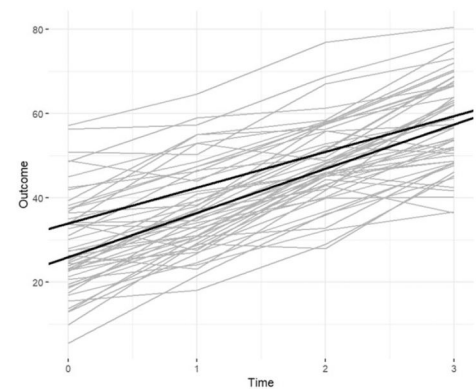
(B)



(C)



(D)



(E)

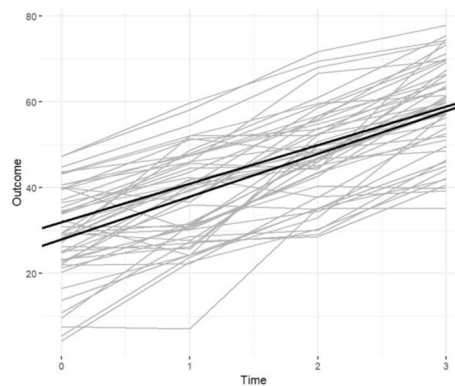


Figure 1. Example simulated two group data with group means differing by (A) 2.5 standard deviations, (B) 2.0 standard deviations, (C) 1.5 standard deviations, (D) 1.0 standard deviations, and (E) 0.5 standard deviations.

All models were fit in Mplus version 8.10 (Muthén & Muthén, 1998-2012) using the SEM and MLM frameworks. The SEM and MLM frameworks have different default starting values. To account for this, we set the starting values for the means of the intercept and slope, the variances of the intercept and slope, as well as the residual variance. The starting values for the means of the intercept and slope for each latent class were different, but similar to the population values to aid estimation in the 2-class GMM.

One-thousand datasets were generated from each population.

7. Results

7.1. Linear Growth Model

The average $-2LL$ for the linear growth model for each simulation condition is reported in Table 2. Additionally, a

Table 2. Mean -2 log-likelihood values for the linear growth model for each simulation condition.

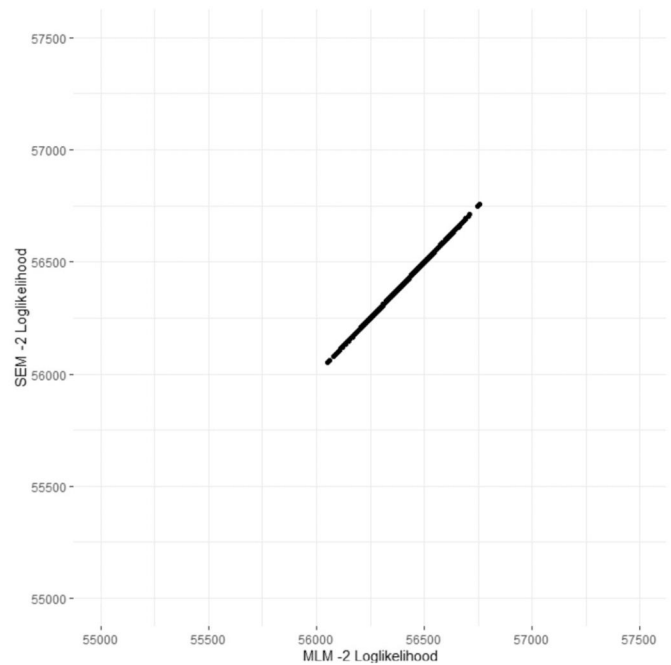
	Structural Equation Model	Multilevel Model
<i>N</i> = 2000		
0.5 Standard Deviations	54,712.92	54,712.92
1.0 Standard Deviations	55,020.25	55,020.25
1.5 Standard Deviations	55,446.57	55,446.57
2.0 Standard Deviations	55,922.82	55,922.82
2.5 Standard Deviations	56,404.07	56,404.07
<i>N</i> = 1500		
0.5 Standard Deviations	41,028.72	41,028.72
1.0 Standard Deviations	41,259.78	41,259.78
1.5 Standard Deviations	41,579.81	41,579.81
2.0 Standard Deviations	41,937.08	41,937.08
2.5 Standard Deviations	42,302.13	42,302.13
<i>N</i> = 1000		
0.5 Standard Deviations	27,354.53	27,354.53
1.0 Standard Deviations	27,508.72	27,508.72
1.5 Standard Deviations	27,722.18	27,722.18
2.0 Standard Deviations	27,960.41	27,960.41
2.5 Standard Deviations	28,201.40	28,201.40
<i>N</i> = 500		
0.5 Standard Deviations	13,678.31	13,678.31
1.0 Standard Deviations	13,755.31	13,755.31
1.5 Standard Deviations	13,861.96	13,861.96
2.0 Standard Deviations	13,981.05	13,981.05
2.5 Standard Deviations	14,099.17	14,099.17

plot of the $-2LL$ values for the linear growth models fit to the simulation condition with $N = 2000$ and a class separation of 2.5 standard deviations is shown in Figure 2, where the values obtained from the MLM framework are featured on the x -axis and the values from the SEM framework are shown on the y -axis. The $-2LL$ values were essentially identical across frameworks, indicating that the estimation of the linear growth model did not depend on the modeling framework. The consistency of the $-2LL$ s was similar for all simulation conditions.

7.2. 2-Class Growth Mixture Model

Next, we examined model fit and parameter estimates for the 2-class GMM, which mapped onto how the data were generated. As a reminder, there were four sample size conditions and five class separation conditions. Class separation affected the ability of the two frameworks to yield the same model fit information and parameter estimates more than sample size. Thus, we focus on our findings with $N = 2000$. Tables for the smaller sample size conditions are found on the second author's website.

When the class means were separated by 2.5 standard deviations, there was a lot of consistency in the model fit and model parameters across the two frameworks. Figure 3 contains six plots to highlight our results. Figure 3A is a histogram showing the difference in $-2LL$ values from the MLM and SEM frameworks. Figures 3B through 3F are scatterplots with values on the x -axis from the MLM estimation and values on the y -axis from the SEM estimation, and vertical and horizontal dashed lines indicating the population value. Figure 3B is the estimate of the latent class mean parameter, indicating the proportion of individuals classified in each latent class (in the logit metric). The next four scatterplots are for the estimates of the latent class means for the intercept and slope. As a reminder, the population

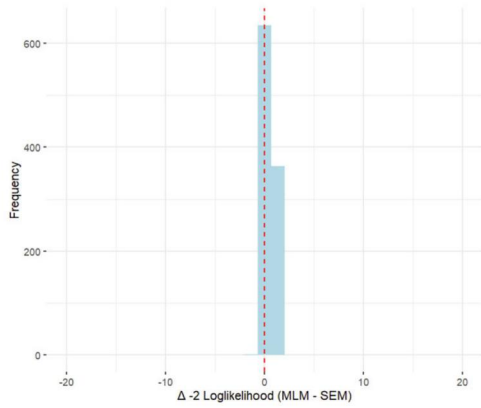
**Figure 2.** Bivariate scatterplot of -2 log-likelihood values from the linear growth model fit in the MLM and SEM frameworks.

means of the intercept and slope were 40 and 7 for the first class, and 20 and 12 for the second class. Examining these plots indicates the MLM and SEM frameworks produced nearly identical results because there was a near perfect association of the $-2LL$ and the key parameter estimates across frameworks. Moreover, the class estimates of the intercept and slope means had low relative bias (see Table 3).

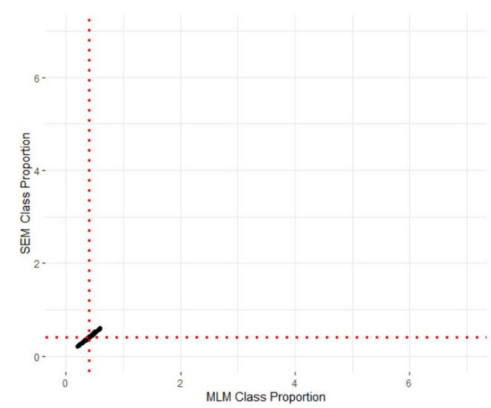
As the class separation decreased, there was greater bias in the parameter estimates and differences in the model fit and parameter estimates emerged between the SEM and MLM frameworks. This can be seen by examining Table 3 and comparing Figures 3, 4, and 5. Table 3 contains mean relative bias values for the latent class means of the intercept and slope from the 2-class linear GMM when estimated in the two frameworks. Figures 4 and 5 mimic the organization structure of Figure 3 (i.e., difference in $-2LL$ and scatterplots of estimates of key parameters) when class separation was 1.5 standard deviations ($N = 2000$) and when the class separation was 0.5 standard deviations ($N = 2000$), respectively. Examining Table 3, we see that the mean relative bias values generally increased as class separation decreased in both frameworks. This was clear when the model was estimated in the MLM framework; however, some relative bias values decreased when class separation was 0.5 standard deviations apart (e.g., Class 1 mean intercept and slope). This may have happened because the GMM may have been unable to identify the two simulated classes. In these cases, the first class may represent the average of the two classes and the second class may represent a few outliers.

Examining and comparing Figures 3, 4, and 5, we see the impact of class separation as there were more values in the scatterplots that were off the main diagonal clustered by the population value as class separation decreased. Thus,

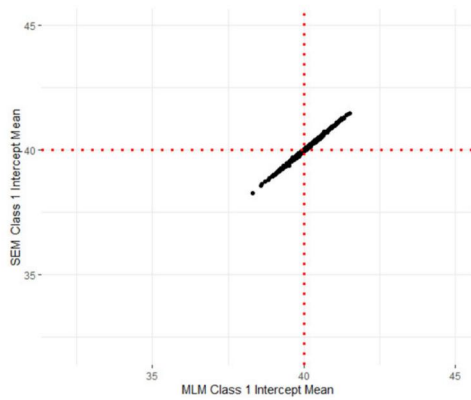
(A)



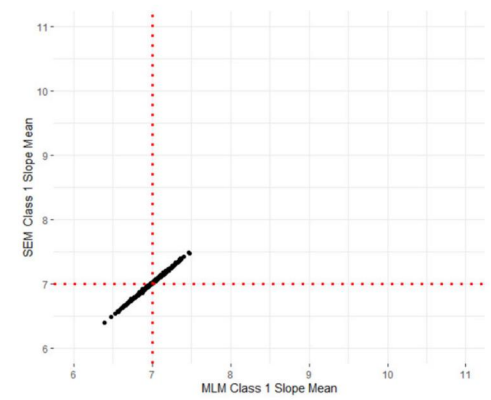
(B)



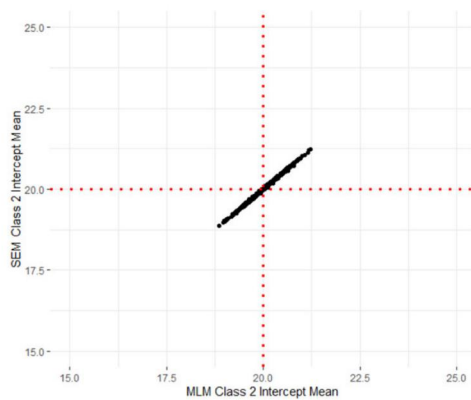
(C)



(D)



(E)



(F)

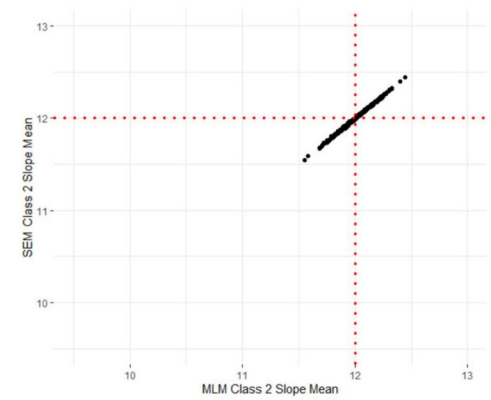


Figure 3. Bivariate scatterplot of estimates from the 2-class linear growth mixture model fit to the data with $N = 2000$ and a class separation of 2.5 standard deviations in the MLM and SEM frameworks for the (A) change in -2 log-likelihood, (B) class mean, (C) class 1 intercept mean, (D) class 1 slope mean, (E) class 2 intercept mean, and (F) class 2 slope mean.

when the classes overlapped more, it was more difficult for the GMM to correctly identify the classes that underlie the data. This result is consistent with prior research (Grimm et al., 2013; Peugh & Fan, 2012; Tofighi & Enders, 2008) that has shown that a high degree of class separation is needed for GMMs to correctly identify the underlying classes.

Even though both frameworks had difficulties correctly estimating the parameters of the GMMs when class separation was low, the scatterplots showed that the MLM and SEM frameworks did not always have the same difficulties. That is, the values in the scatterplots that lie off the main diagonal are datasets where the MLM and SEM frameworks

Table 3. Relative bias of the latent class means of the intercept and slope from the 2-class linear growth mixture model in the SEM and MLM frameworks.

Class Separation	Structural Equation Model		Multilevel Model	
	Class 1	Class 2	Class 1	Class 2
2.5 Standard Deviations				
Intercept mean	0.01%	0.10%	0.04%	−0.10%
Slope mean	−0.09%	0.07%	−0.19%	0.09%
2.0 Standard Deviations				
Intercept mean	−3.11%	4.67%	0.16%	0.23%
Slope mean	4.28%	−2.69%	0.16%	0.02%
1.5 Standard Deviations				
Intercept mean	−9.79%	3.01%	−0.25%	0.54%
Slope mean	15.43%	−12.69%	0.04%	−0.34%
1.0 Standard Deviations				
Intercept mean	−7.97%	−6.12%	1.54%	−1.27%
Slope mean	14.52%	−13.38%	1.00%	−2.42%
0.5 Standard Deviations				
Intercept mean	−2.45%	−15.27%	5.03%	7.24%
Slope mean	7.99%	−8.69%	−0.21%	−0.45%

arrived at different solutions. The scatterplots have values above this main diagonal indicating datasets where the MLM framework appeared to properly estimate key parameters and the SEM framework appeared to have trouble estimating these parameters. Similarly, the scatterplots also show values to the right of this main diagonal indicating datasets where the SEM framework appears to properly estimate the parameter and the MLM framework appears to have trouble estimating these parameters. The mean relative bias values and the scatterplots of the parameter estimates indicate that the MLM framework tended to obtain more accurate estimates compared to the SEM framework as class separation decreased; however, this wasn't the case across all datasets within a simulation condition.

7.3. 3-Class Growth Mixture Model and Model Comparison

A 3-class GMM was estimated to examine whether model selection, based on the SABIC, varied as a function of the modeling framework. The average SABIC values across replications and the percentage of replications in which the 2-class model had the lowest SABIC are shown in Table 4. Overall, the 2-class model tended to have a lower SABIC than both the growth model and the 3-class GMM under simulation conditions with high class separation and large sample size. For example, when class separation was 2.5 standard deviations and sample size was $N=1000$, the 2-class model had the lowest BIC in 55.7% of the data sets in the SEM framework and 57.0% of the data sets in the MLM framework. The differences in SABIC values for the 2-class model across frameworks were modest, with the SEM framework generally yielding lower SABIC values than the MLM framework. This was, at least partially, due to how sample size is determined in Mplus for these calculations. Mplus considers sample size for information criteria as the number of rows in the dataset, which is the actual sample size in the SEM framework, but a combination of sample size and number of time points per person in the MLM framework. Thus, the difference in how the SABIC is calculated contributed to the differences in the SABIC values

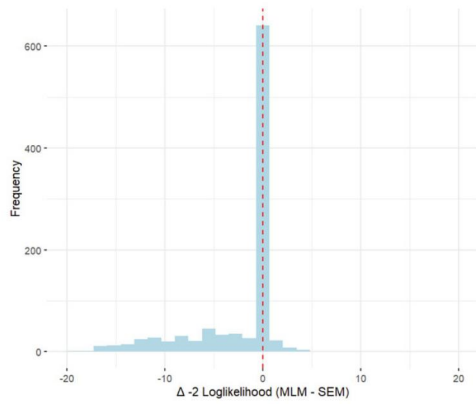
across frameworks and the small differences in the percent of simulations where the 2-class model had the lowest SABIC. When class separation was less than 2 standard deviations, the classes were difficult to differentiate based on the SABIC as the growth (1-class) model tended to have the lowest SABIC values.

8. Discussion

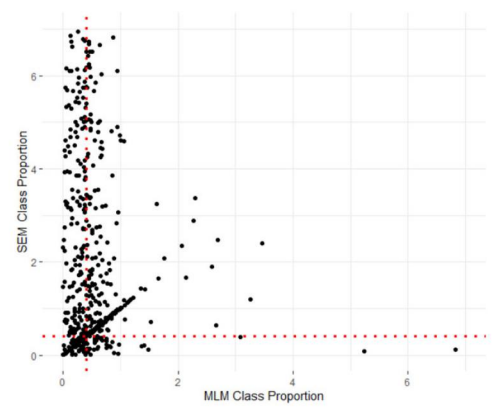
Growth models and GMMs can be specified and estimated in both the SEM and the MLM frameworks within Mplus. These modeling frameworks produce identical model fit statistics and parameter estimates when estimating linear growth models (Willett & Sayer, 1994); however, discrepancies between frameworks in estimating linear GMMs arose under certain conditions. When class separation was large, the MLM and SEM frameworks produced identical or nearly identical parameter estimates and model fit information. Moreover, the estimates were accurate, showing low relative bias. Inaccurate parameter estimates and parameter discrepancies between frameworks arose as class separation decreased. Overall, the parameter estimates from the MLM framework tended to be more accurate; however, the SEM framework did outperform the MLM framework for some simulated datasets. When the classes were very close together, 0.5 standard deviations apart, the estimates from the two frameworks tended to be different, and neither framework accurately recovered all of the population parameters. This suggests that both the MLM and SEM frameworks performed well when estimating linear GMMs under conditions of large class separation following previous research (Grimm et al., 2013; Peugh & Fan, 2012; Tofighi & Enders, 2008), but the performance of the SEM and MLM frameworks for estimating linear GMMs declined as class separation decreased—oftentimes in different ways. Additional plotting confirmed that instances of poor parameter recovery were consistently associated with poor model fit (supporting plots are found on the second author's website).

To better understand why the discrepancies in the parameter estimates and model fit arose between frameworks, we modified specific parameters of the estimation routines in an attempt to optimize consistency between frameworks. While these modifications aimed to make the settings within our control equal, inherent differences in estimation routines remained. This research was done using the weight data from the ECLS-K discussed above. First, we increased the number of random starts. Random starts can aid model estimation by helping to avoid local maxima in the likelihood function. We increased the number of initial stage random starts to 1,000, the number of final stage optimizations to 20, and the number of initial stage iterations to 10. The parameter estimates remained different between frameworks. The model fit information did not change in the SEM specification and estimation, whereas the MLM model fit information got slightly worse. The MLM model fit information (i.e., $-2LL$) remained better than the model fit information from the SEM framework.

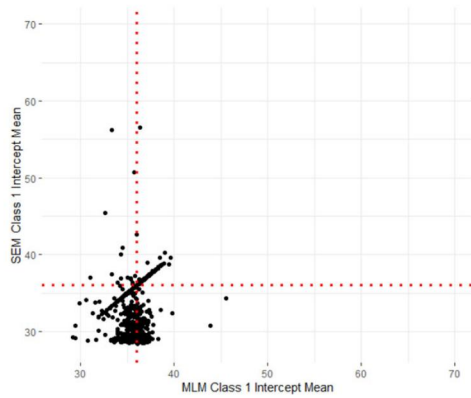
(A)



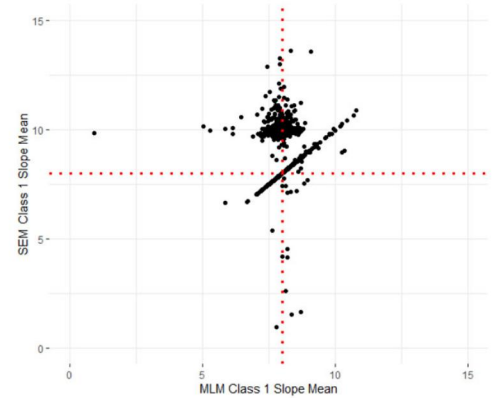
(B)



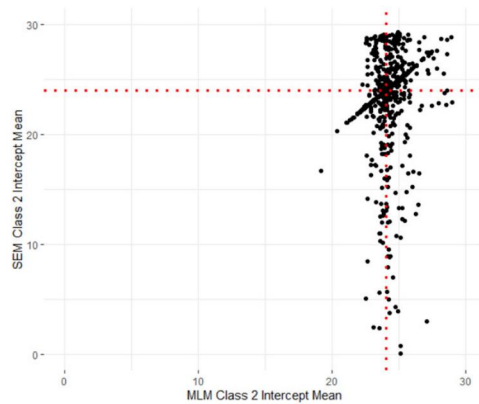
(C)



(D)



(E)



(F)

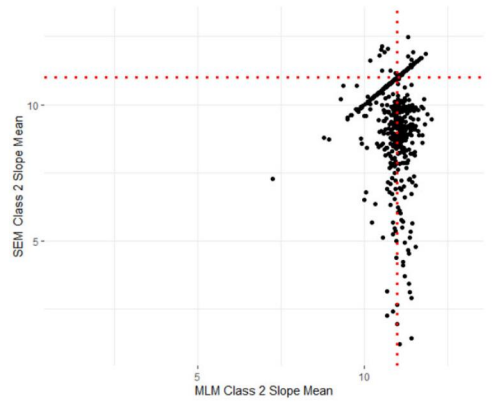
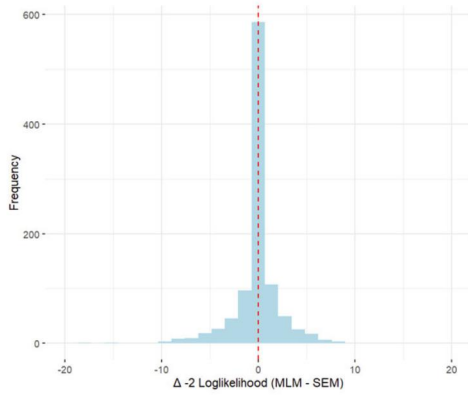


Figure 4. Bivariate scatterplot of estimates from the 2-class linear growth mixture model fit to the data with $N = 2000$ and a class separation of 1.5 standard deviations in the MLM and SEM frameworks for the (A) change in -2 log-likelihood, (B) class mean, (C) class 1 intercept mean, (D) class 1 slope mean, (E) class 2 intercept mean, and (F) class 2 slope mean.

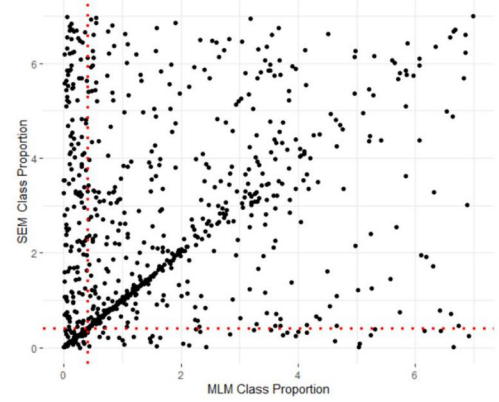
Next, we aligned the convergence criteria and step sizes. The convergence criteria differed across frameworks, with the SEM framework using slightly stricter thresholds. To align them, we adjusted the MLM framework's convergence criteria to match those used in the SEM framework by setting the convergence criteria to $0.100e-05$, the convergence

criteria for log likelihood change to $0.100e-06$, and the convergence criteria for relative log likelihood change to $0.100e-06$. Interestingly, the MLM framework model fit worsened after adjusting the convergence criteria. The SEM framework model fit remained stable, resulting in the SEM-based GMM showing a better model fit than the MLM-based

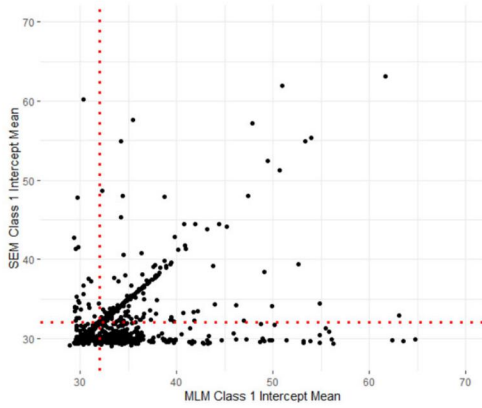
(A)



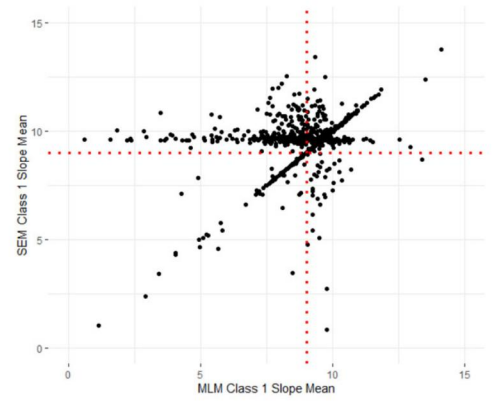
(B)



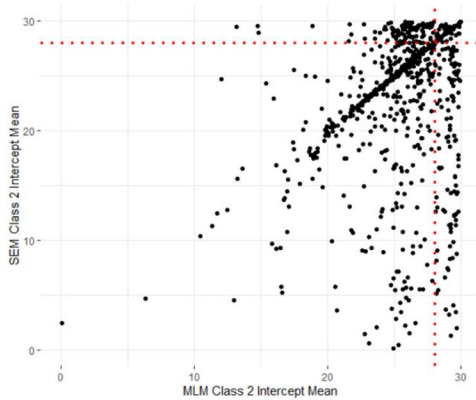
(C)



(D)



(E)



(F)

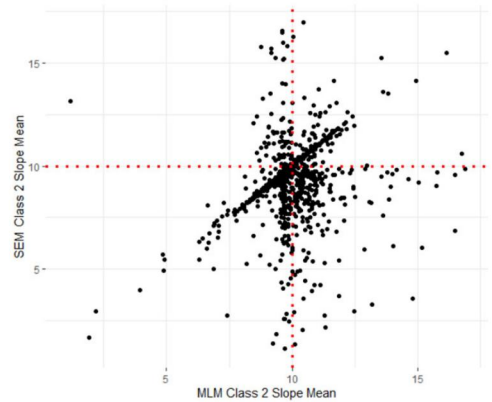


Figure 5. Bivariate scatterplot of estimates from the 2-class linear growth mixture model fit to the data with $N = 2000$ and a class separation of 0.5 standard deviations in the MLM and SEM frameworks for the (A) change in -2 log-likelihood, (B) class mean, (C) class 1 intercept mean, (D) class 1 slope mean, (E) class 2 intercept mean, and (F) class 2 slope mean.

GMM. The parameter estimates and remaining model fit statistics remained different across frameworks.

Finally, we addressed a difference in the M-step convergence criteria. The SEM framework had a more constrained convergence criteria for the M-step when estimating categorical latent variables. We matched this by setting the MLM framework's M-step convergence criterion to $0.1e-05$

and re-ran the models. The differences in the parameter estimates remained and the model fit was nearly identical to the previous models, where we adjusted the likelihood-based convergence criteria. Thus, the differences remained when estimating the 2-class GMM across the two frameworks when we attempted to align the parameters related to estimation. Thus, even though the same model is being

Table 4. Mean sample size adjusted bayesian information criterion values for the linear growth model, two-class and three-class growth mixture models.

	Structural Equation Model				Multilevel Model			
	Growth	2-class	3-class	%	Growth	2-class	3-class	%
<i>N</i> = 2,000								
0.5 SD	54,758.5	54,777.4	54,797.7	0	54,766.8	54,816.8	54,814.3	0
1.0 SD	55,065.9	55,084.41	55,105.0	0	55,074.2	55,096.4	55,122.8	0
1.5 SD	55,492.2	55,505.4	55,528.6	3.7	55,500.5	55,515.24	55,545.5	2.0
2.0 SD	55,968.4	55,948.2	55,972.6	52.3	55,976.7	55,960.4	55,988.8	51.0
2.5 SD	56,449.7	56,352.2	56,374.6	55.9	56,458.0	56,365.4	56,391.6	56.3
<i>N</i> = 1,500								
0.5 SD	41,072.6	41,090.6	41,115.2	0	41,080.9	41,103.0	41,132.8	0
1.0 SD	41,303.7	41,321.5	41,348.1	0	41,312.0	41,333.4	41,365.0	0
1.5 SD	41,623.7	41,637.4	41,663.2	1.6	41,632.0	41,648.0	41,680.0	0.4
2.0 SD	41,981.0	41,970.4	41,992.5	49.2	41,989.3	41,982.1	42,009.2	42.6
2.5 SD	42,302.1	42,211.9	42,212.1	49.4	42,354.3	42,290.7	42,310.5	54.7
<i>N</i> = 1,000								
0.5 SD	27,396.0	27,412.9	27,423.7	0	27,404.3	27,425.1	27,440.8	0
1.0 SD	27,550.2	27,566.8	27,582.7	0	27,558.5	27,578.9	27,599.7	0
1.5 SD	27,763.6	27,777.4	27,791.6	0.6	27,771.9	27,788.6	27,808.0	0.1
2.0 SD	28,001.9	28,000.1	28,022.4	32.3	28,010.2	28,011.3	28,039.1	22.9
2.5 SD	28,242.9	28,202.1	28,215.4	55.7	28,251.2	28,214.8	28,232.0	57.0
<i>N</i> = 500								
0.5 SD	13,719.8	13,736.4	13,742.5	0	13,723.9	13,742.4	13,759.5	0
1.0 SD	13,796.8	13,813.4	13,818.7	0	13,800.9	13,819.4	13,835.6	0
1.5 SD	13,903.4	13,918.8	13,922.0	0.1	13,907.6	13,924.3	13,938.4	0.1
2.0 SD	14,022.5	14,030.8	14,037.4	8.0	14,026.7	14,036.0	14,053.3	5.9
2.5 SD	14,140.6	14,129.1	14,135.3	45.3	14,135.2	14,151.7	14,151.7	45.6

Notes: The % column indicates the percentage of simulated data sets where the 2-class model had the lowest BIC value for each simulation condition.

estimated in the two frameworks, the estimation approach differences (EM versus EM with quadrature) lead to a different navigation of the log-likelihood surface, which appears to lead to different propensities to settle on local maxima.

9. Limitations

This study was limited to a specific set of simulation conditions and the use of Mplus. Note, the growth factors were uncorrelated in the data generating model but the growth factors were allowed to correlate in model's estimation. Having correlated growth factors in the population may improve or reduce parameter recovery; however, our primary focus was to examine the comparative performance of the two frameworks, and we do not expect substantially different results if the growth factors were correlated within each group in the population. Next, we focused on linear GMMs with two classes in the population that only differed in the means of the growth factors. Future research should expand on these conditions by exploring GMMs with mean and variance differences between latent classes, nonlinear growth trajectories, populations with more than two classes, unbalanced data, alternative estimation techniques (e.g., MCMC), and different statistical software (e.g., how does estimation in the lcmm package (Proust-Lima et al., 2017) compare to estimation in Mplus). Future work in this area is important to further understand the strengths and limitations of the SEM and MLM frameworks for estimating GMMs.

10. Concluding Remarks

We explored when and how differences in model fit and parameter estimates of GMMs emerge when estimated in

the SEM and MLM frameworks in Mplus. Through simulation analyses, we found that both frameworks yield comparable parameter estimates when class separation was large; however, parameter estimation accuracy worsened in both frameworks and discrepancies between the two frameworks arose as class separation decreased. The MLM framework seemed to outperform the SEM framework in the majority of simulations where discrepancies were found; however, there were simulated datasets where the SEM framework outperformed the MLM framework. Given these results, our recommendation is to focus on using the MLM framework for linear GMMs, but to re-estimate the final set of models in the SEM framework to confirm model selection and parameter estimation. If model selection and parameter estimation differ, we recommend referring to the $-2LL$ and report the model with the lowest $-2LL$ for each model type (i.e., 2-class model with means allowed to vary over latent classes).

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